SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO2 ASSESSMENT TOOL 1 – Scenario-Based Requirement Analysis

Course Code	CSA10 / CSA1063	**Course Title**	Software Engineering
CO Assessed	CO2	**Assessment**	Scenario-Based Requirement Analysis
Weightage	20%	**Total Marks**	25
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Date	26 October 2026	**Duration**	60 minutes

CO2 Statement: Perform detailed requirements analysis, design scalable architectures, and prioritize features with a focus on cross-disciplinary skills

Code of Conduct

I V.Shashank sai certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Shashank sai

1. SCENARIO STUDY AND ANALYSIS

Project Title: QuickBite – Campus Canteen Pre-Ordering System

1.1 Business Domain and Problem Context

Domain: Campus food service and digital ordering. A typical college canteen serves 2,000 to 3,000 students and staff every day, but most of the demand falls into a 30 to 45 minute lunch break. Customers stand in long queues, orders are taken verbally or on paper, payments are made in cash or through separate UPI scans, and the kitchen learns what to cook only when the crowd arrives. Many students skip meals or return late to class, while the canteen loses sales and wastes food. QuickBite lets a customer choose items from a live menu, pick a pickup time slot, pay online and collect the food with a QR code. The kitchen receives a time-ordered queue, and the manager sees demand and sales in real time.

1.2 Existing Challenges

- Long queues: students lose 15 to 20 minutes of a short break waiting to order and pay.
- Unknown demand: the kitchen cannot predict how much to prepare, causing both shortages and food wastage.
- Stale menu information: sold-out items are discovered only at the counter.
- Order errors: verbal or hand-written orders lead to wrong or missed items and disputes.
- Payment problems: cash handling, shortage of change and manual reconciliation of UPI payments.
- No order history or feedback: complaints are given verbally and never followed up.
- Little management insight: sales, peak hours and popular items are not recorded for decisions.

1.3 System Objectives

1. Let customers browse a live menu and place orders from a phone or computer without queueing.
2. Spread demand across pickup time slots so that the kitchen is never overloaded.
3. Accept secure online payments, wallet top-ups and automatic refunds.
4. Give kitchen staff a live, time-ordered order queue and fast QR-based handover.
5. Show the manager real-time sales, stock and demand data, and collect customer feedback.
6. Keep all data secure, available during canteen hours and auditable.

1.4 Stakeholders

Stakeholder	Type	Needs from the system
Customer (student / faculty / staff)	Primary user	Browse the menu, order without queueing, pay online, track the order, collect it quickly and give feedback.
Canteen Staff (kitchen / counter)	Primary user	See incoming orders in pickup-time order, update order status and verify orders at handover.
Canteen Manager	Primary user	Maintain menu, prices, stock and slots; view sales and demand reports; respond to feedback.
System Administrator	Primary user	Manage accounts, security, backups and the audit log.
Institution / Canteen contractor	Sponsor	Shorter queues, higher sales, less wastage, low hosting cost and data privacy compliance.
Finance / accounts office	Secondary user	Read-only sales and refund reports for reconciliation.
Payment gateway and notification providers	External systems	Process payments and refunds; deliver email, SMS and push notifications.

2. FUNCTIONAL AND NON-FUNCTIONAL REQUIREMENTS

2.1 Functional Requirements

36 functional requirements are grouped by module and prioritised with MoSCoW (M = Must, S = Should, C = Could).

ID	Requirement	Pri.
User and access management		
FR-01	The system shall let a new user register with name, institutional email, phone number and role request (customer or staff).	M
FR-02	The system shall verify the email address or phone number with a one-time code before activating an account.	M
FR-03	The system shall authenticate users by email and password, with optional two-factor authentication for staff and administrators.	M
FR-04	The system shall enforce role-based access so each role sees only its permitted functions and data.	M
FR-05	The administrator shall be able to create, edit, suspend and deactivate user accounts, including canteen staff.	M
FR-06	The system shall let users reset a forgotten password through a time-limited email link.	M
Menu and stock management		
FR-07	The manager shall be able to add, edit and remove menu items with name, description, category, price, photo and veg / non-veg mark.	M
FR-08	The manager shall be able to set item availability and daily stock; an item shall show as Sold Out when its stock reaches zero.	M
FR-09	The manager shall be able to define time-based menus (breakfast, lunch, snacks).	S
FR-10	The manager shall be able to mark daily specials and apply discounts or offers.	C
FR-11	A customer shall be able to browse, search and filter the menu by category, diet type and price.	M
Cart, slots and ordering		
FR-12	A customer shall be able to add items to a cart, change quantities and add special instructions.	M
FR-13	A customer shall be able to choose a pickup time slot from the available slots.	M

ID	Requirement	Pri.
FR-14	The system shall limit the number of orders per slot to a capacity set by the manager (default 100 orders per 10-minute slot).	S
FR-15	The system shall calculate the order total with taxes and discounts and show it before the customer confirms.	M
FR-16	The system shall confirm an order only after successful payment and issue a unique order number and QR code.	M
FR-17	A customer shall be able to cancel an order while its status is Received (kitchen has not started).	S
FR-18	A customer shall be able to reorder a previous order from order history.	C
Payments and refunds		
FR-19	The system shall accept online payment by UPI, cards and net banking through a payment gateway.	M
FR-20	The system shall provide a wallet that a customer can top up and use to pay for orders.	S
FR-21	The system shall automatically refund cancelled or rejected orders to the original payment method or wallet.	M
FR-22	The system shall generate a digital receipt for every paid order.	S
Kitchen and order fulfilment		
FR-23	Canteen staff shall see a live queue of paid orders sorted by pickup time.	M
FR-24	Canteen staff shall be able to accept or reject an order, giving a reason when rejecting.	M
FR-25	Canteen staff shall be able to update the order status: Received, Preparing, Ready, Collected.	M
FR-26	Canteen staff shall be able to scan the order QR code to verify and hand over an order.	M
FR-27	The system shall mark an order as No-show if it is not collected within 30 minutes after its slot ends.	S
Notifications and feedback		
FR-28	The system shall send in-app, email or SMS alerts for order confirmation, order ready, rejection and cancellation.	M
FR-29	The system shall remind a customer 10 minutes before the pickup slot starts.	S
FR-30	A customer shall be able to rate and review a collected order and its items.	S
FR-31	The manager shall be able to view feedback and post a reply to a review.	C
Reporting and administration		
FR-32	The manager shall see a dashboard with today’s orders, revenue, peak hours and most popular items.	M
FR-33	The system shall generate daily, weekly and monthly sales reports as PDF and CSV.	S
FR-34	The system shall alert the manager when an item’s stock falls below a set level.	S
FR-35	The administrator shall be able to back up and restore data and view the audit log.	M
FR-36	The system shall write an audit log entry for each create, update, delete, login, price change and refund event.	M

Summary: 23 Must, 10 Should, 3 Could.

2.2 Non-Functional Requirements

Each non-functional requirement carries a measurable target so that it can be tested.

ID	Category	Requirement	Measure / Target
NFR-01	Performance	Pages shall load quickly under normal load, including the lunch-hour peak.	95% of pages < 3 s with 500 concurrent users

ID	Category	Requirement	Measure / Target
NFR-02	Performance	Menu search, order placement and kitchen queue updates shall respond fast.	< 2 s for 5,000 menu / order records; queue refresh < 5 s
NFR-03	Security	Passwords stored hashed and salted; all traffic encrypted.	bcrypt or stronger; HTTPS (TLS 1.2+)
NFR-04	Security	Access follows RBAC; sessions expire; card details are never stored; common web attacks resisted.	30 min idle logout; PCI DSS compliant gateway; no OWASP Top-10 findings
NFR-05	Reliability	The system shall run without failure in normal use.	MTBF ≥ 500 h
NFR-06	Reliability	Paid orders and payment records shall not be lost on failure.	Daily backup plus transaction log; recovery point 1 h, recovery time 2 h
NFR-07	Availability	Available during canteen operating hours.	99.5% uptime from 7 AM to 7 PM excluding planned maintenance
NFR-08	Usability	A new customer shall place an order without training; accessible and responsive.	Order placed in < 2 min; WCAG 2.1 AA; 360–1920 px
NFR-09	Scalability	The system shall grow with more users without redesign.	5,000 users and 600 orders/hour at peak now; 20,000 users by adding servers
NFR-10	Maintainability	Code shall be modular, tested and documented.	≥ 70% unit-test coverage; REST API documented
NFR-11	Portability	The web client shall work on current browsers and mobile browsers.	Latest 2 versions of Chrome, Edge, Firefox, Safari; Android and iOS browsers
NFR-12	Auditability	Changes to prices, orders and refunds shall be traceable.	100% of changes logged with user and time
NFR-13	Privacy	Personal data used only for canteen service and academic purposes.	Compliant with DPDP Act, 2023

3. ACTORS AND USE CASES

3.1 Actors

Actor	Type	Description
Customer	Primary	Student, faculty or staff member who browses the menu, orders, pays, collects the food and gives feedback.
Canteen Staff	Primary	Kitchen or counter employee who accepts and prepares orders, updates status and hands over food.
Canteen Manager	Primary	Runs the canteen: menu, stock, slots, reports and feedback (specialises Canteen Staff).
Administrator	Primary	Manages accounts, backups and the audit log.
Payment Gateway	Secondary	External service (for example Razorpay or a UPI provider) that processes payments and refunds.
Notification Service	Secondary	External email, SMS and push provider that delivers notifications.

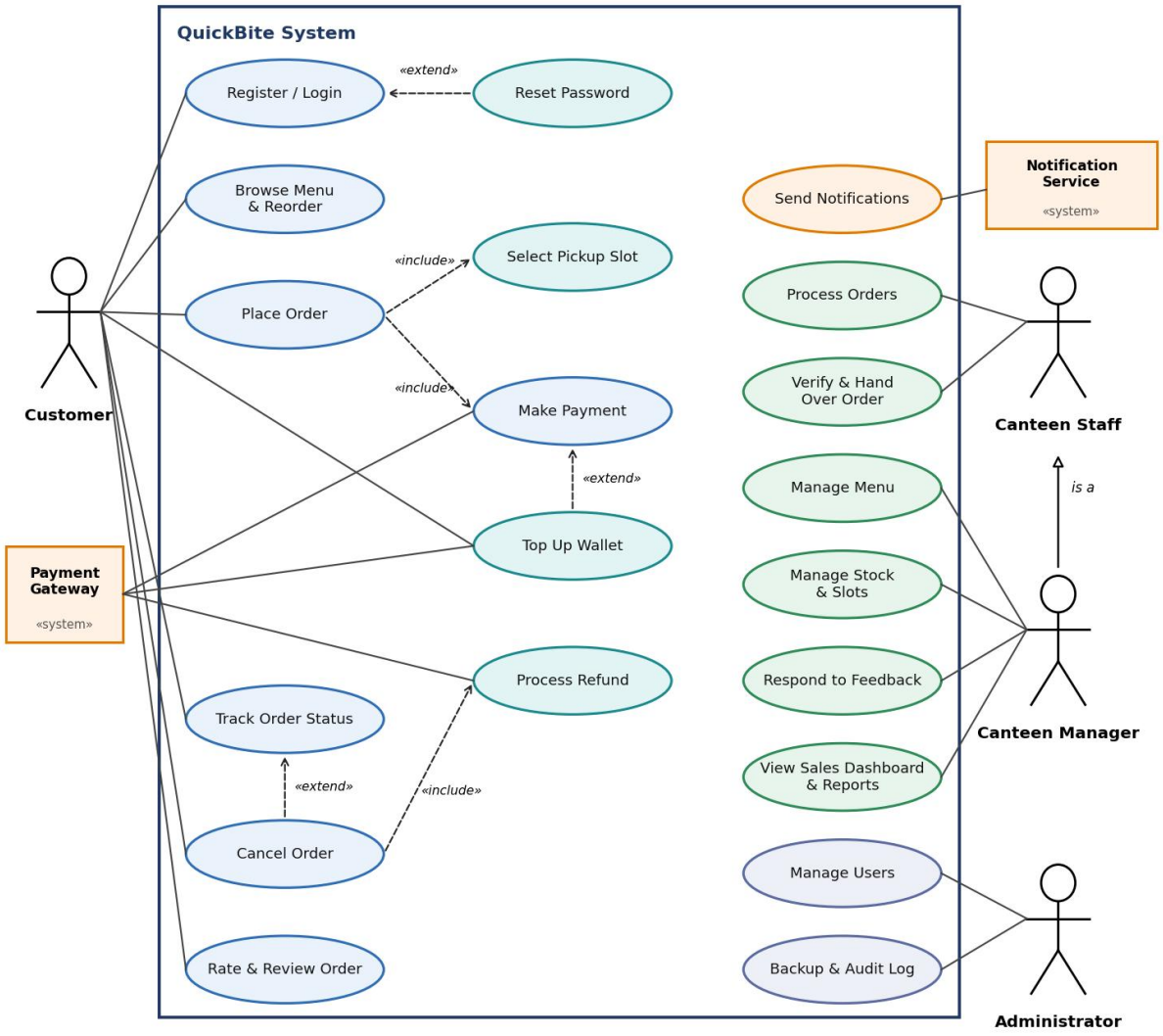
3.2 Major Use Cases by Actor

ID	Use Case	Actor(s)	Related FR
UC-01	Register / Login (extended by UC-20)	All human actors	FR-01 – FR-04
UC-02	Manage Users	Administrator	FR-05
UC-03	Manage Menu	Canteen Manager	FR-07 – FR-10
UC-04	Manage Stock & Slots	Canteen Manager	FR-08, FR-14, FR-34
UC-05	Browse Menu & Reorder	Customer	FR-11, FR-18
UC-06	Place Order (includes UC-07, UC-18)	Customer	FR-12, FR-15, FR-16
UC-07	Make Payment (extended by UC-17)	Customer, Payment Gateway	FR-19, FR-20, FR-22
UC-08	Cancel Order (extends UC-09, includes UC-19)	Customer, Payment Gateway	FR-17
UC-09	Track Order Status	Customer	FR-28, FR-29
UC-10	Rate & Review Order	Customer	FR-30
UC-11	Process Orders	Canteen Staff	FR-23 – FR-25
UC-12	Verify & Hand Over Order	Canteen Staff	FR-26, FR-27
UC-13	Send Notifications	Notification Service (system-triggered)	FR-28, FR-29
UC-14	View Sales Dashboard & Reports	Canteen Manager	FR-32, FR-33
UC-15	Respond to Feedback	Canteen Manager	FR-31
UC-16	Backup & Audit Log	Administrator	FR-35, FR-36
UC-17	Top Up Wallet (extends UC-07)	Customer, Payment Gateway	FR-20
UC-18	Select Pickup Slot (included in UC-06)	Customer	FR-13, FR-14
UC-19	Process Refund (included in UC-08)	Payment Gateway (system-triggered)	FR-21
UC-20	Reset Password (extends UC-01)	All human actors	FR-06

3.3 Use Case Diagram

The diagram follows UML notation: actors outside the system boundary, use cases as ovals, solid lines for associations, a solid hollow arrow for generalisation and dashed arrows for «include» and «extend» relationships.

QuickBite – Use Case Diagram



Dashed arrows: «include» / «extend». Hollow arrow: Canteen Manager is a kind of Canteen Staff. Shaded boxes: external systems. All human actors log in through Register / Login.

4. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

4.1 Introduction

Purpose: This SRS defines what QuickBite must do and how well it must do it, giving customers, canteen staff, the manager, administrators, developers and testers one agreed baseline. **Audience:** canteen management, kitchen and counter staff, administrators, developers and testers. **Conventions:** requirements are numbered FR-xx and NFR-xx, use cases UC-xx; priorities follow MoSCoW. **References:** IEEE 830 / ISO/IEC/IEEE 29148; Sommerville, Software Engineering; Pressman, Software Engineering: A Practitioner's Approach.

4.2 Problem Description

Customers queue to order and pay during a short break, the kitchen cannot predict demand, sold-out items are discovered late and payments are reconciled by hand. QuickBite is a single web platform where a customer views a live menu, selects a pickup slot, pays online and collects food with a QR code, while the kitchen works from a time-ordered queue and the manager sees sales and stock in real time.

4.3 Scope of the System

In scope	Out of scope (this release)
• Registration, login and role-based access • Menu, stock and time-based menus • Cart, pickup slots and same-day ordering • Online payment, wallet and refunds • Kitchen queue, order status and QR handover • Notifications and order tracking • Feedback and reviews • Dashboards, reports, backup and audit log	• Home or hostel delivery and delivery tracking • Multiple canteens or outside restaurants • Cash payment and counter billing (POS) integration • Supplier purchasing and full inventory management • Loyalty points and advance (future-day) orders • Native mobile apps (responsive web only)

4.4 Use Case Descriptions

UC-01 Register / Login

Field	Description
Actor	Customer, Canteen Staff, Canteen Manager, Administrator
Precondition	User has an institutional email address and a mobile number.
Main flow	1. User chooses Register or Login. 2. For registration, user enters name, email, phone number and role; system sends a verification code. 3. User verifies the code; account becomes active (staff accounts are approved by the administrator). 4. User enters email and password (and 2FA code if enabled). 5. System validates credentials and opens the role's home screen.
Alternate flows	A1: Wrong password – error shown; after 5 failures the account locks for 15 min. A2: Forgotten password – user requests a reset link by email (UC-20).
Postcondition	User has an authenticated session with role-based permissions.

UC-03 Manage Menu

Field	Description
Actor	Canteen Manager
Precondition	Manager is logged in.
Main flow	1. Manager opens Menu Management. 2. Adds or edits an item with name, description, category, price, veg / non-veg mark and photo. 3. Assigns the item to a time-based menu and sets daily stock. 4. Saves; the item is visible to customers immediately. 5. System writes the change to the audit log (FR-36).
Alternate flows	A1: Price missing or not greater than zero – system rejects the entry. A2: Item is in an active order – removal is blocked; manager marks it Unavailable instead.
Postcondition	The updated menu is published and the change is logged.

UC-06 Place Order

Field	Description
Actor	Customer (Payment Gateway participates through UC-07)
Precondition	Customer is logged in; at least one item is available and a pickup slot is open.

Field	Description
Main flow	1. Customer adds items to the cart with quantity and special instructions. 2. Customer selects a pickup slot. 3. System shows the total with taxes and discounts. 4. Customer confirms and pays (includes UC-07). 5. System creates the order with a unique order number and QR code. 6. System notifies the customer and the kitchen (UC-13).
Alternate flows	A1: An item sells out while in the cart – system removes it and asks the customer to review. A2: Chosen slot is full – system offers the nearest free slot. A3: Payment fails – no order is created and held stock is released.
Postcondition	A confirmed, paid order exists in the kitchen queue.

UC-07 Make Payment

Field	Description
Actor	Customer, Payment Gateway
Precondition	Cart is confirmed and the total is greater than zero.
Main flow	1. Customer selects UPI, card, net banking or wallet. 2. System redirects to the gateway (or debits the wallet). 3. Gateway returns a success status. 4. System records the payment and issues a digital receipt.
Alternate flows	A1: Payment fails or times out – customer may retry within 5 minutes while items are held, after which they are released. A2: Wallet balance is insufficient – customer tops up the wallet or chooses another method.
Postcondition	Payment is recorded and the order can be confirmed.

UC-17 Top Up Wallet

Field	Description
Actor	Customer, Payment Gateway
Precondition	Customer is logged in; this use case extends UC-07 when the wallet balance is low or when the customer chooses Add Money.
Main flow	1. Customer selects Top Up Wallet and enters an amount. 2. System redirects to the gateway for payment. 3. Gateway confirms the payment. 4. System credits the wallet and records the transaction.
Alternate flows	A1: Payment fails – wallet balance is unchanged and the customer may retry. A2: Amount is outside the allowed limits – system rejects it with a message.
Postcondition	Wallet balance is increased and the transaction is logged.

UC-08 Cancel Order

Field	Description
Actor	Customer, Payment Gateway
Precondition	Order status is Received (the kitchen has not started).

Field	Description
Main flow	1. Customer opens Track Order Status and selects Cancel. 2. System checks that the status is still Received. 3. Customer confirms the cancellation. 4. System cancels the order and releases the slot and stock. 5. System refunds the amount to the original method or wallet (UC-19, FR-21). 6. System notifies the customer (UC-13).
Alternate flows	A1: Status is Preparing or later – the Cancel option is disabled. A2: Refund fails – the case is flagged to the manager and retried.
Postcondition	Order is Cancelled and the refund is recorded in the audit log.

UC-11 Process Orders

Field	Description
Actor	Canteen Staff
Precondition	Staff member is logged in and paid orders exist.
Main flow	1. Staff opens the live queue sorted by pickup time. 2. Staff reviews an order and accepts it. 3. Staff sets the status to Preparing. 4. Staff marks the order Ready; system notifies the customer (UC-13).
Alternate flows	A1: An item is unavailable – staff rejects the order with a reason and a refund is issued automatically. A2: Orders in a slot exceed capacity – system warns the manager.
Postcondition	Order status is updated and visible to the customer.

UC-12 Verify & Hand Over Order

Field	Description
Actor	Canteen Staff
Precondition	Order status is Ready.
Main flow	1. Customer shows the order QR code. 2. Staff scans it; system validates the order number, payment and status. 3. Staff hands over the food and marks the order Collected.
Alternate flows	A1: QR code is invalid or already used – handover is blocked. A2: Order not collected within 30 minutes after the slot – system marks it No-show; refund follows canteen policy.
Postcondition	Order is closed as Collected (or No-show) and logged.

4.5 Data Requirements

Entity	Key attributes	Relationships
User	user_id (PK), name, email (unique), phone, password_hash, role, status	A user places many orders
MenuItem	item_id (PK), name, description, category, price, veg_flag, photo_path, stock_qty, available	Many order items
PickupSlot	slot_id (PK), slot_date, start_time, end_time, capacity	Many orders
Order	order_id (PK), user_id (FK), slot_id (FK), order_no (unique), qr_code, total_amount, status, created_at	One user, one slot, many order items
OrderItem	orderitem_id (PK), order_id (FK), item_id (FK), quantity, unit_price, instructions	One order, one menu item

Entity	Key attributes	Relationships
Payment	payment_id (PK), order_id (FK), method, gateway_ref, amount, status, paid_at	One order may have refunds
Wallet / WalletTxn	wallet_id (PK), user_id (FK), balance; txn_id (PK), type, amount, created_at	A wallet has many transactions
Feedback	feedback_id (PK), order_id (FK), user_id (FK), rating, comment, manager_reply	One feedback per order
AuditLog	log_id (PK), user_id (FK), action, entity, timestamp, ip_address	Written on every create, update, delete, login, refund

- Email and order number are unique. Prices on an order are copied into OrderItem, so later price changes never alter past orders; nothing is overwritten.
- Order and payment records are kept for 5 years for accounting; audit logs for 2 years. User accounts are deactivated, not deleted, so financial records stay intact.
- Menu photos are limited to 1 MB (JPG or PNG). Estimated load is about 2,500 orders per day with 600 orders per hour at peak, giving about 2 GB of storage per year.

4.6 External Interface Requirements

Interface	Requirement
User interface	Responsive web interface (mobile-first) with a bottom navigation bar for customers and a role-based home screen for staff. Main screens: login, menu, cart, slot selection, payment, order tracking, kitchen queue, manager dashboard, reports, admin console. Keyboard navigation and screen-reader labels (WCAG 2.1 AA).
Hardware	Customers: any smartphone or computer with a modern browser. Counter: tablet or PC with a camera for QR scanning; kitchen: a display screen. Server: Linux VM with at least 4 vCPU, 8 GB RAM and 100 GB storage.
Software	PostgreSQL database; object storage or file system for menu photos; payment gateway API (REST/JSON with webhooks); SMTP and SMS provider; PDF generator for reports; optional LDAP/OAuth directory to verify institutional emails.
Communication	All client-server traffic over HTTPS (TLS 1.2+). Internal REST API with JSON under /api/v1 protected by session tokens. Live order updates over WebSocket or server-sent events. Notifications sent asynchronously through a queue.

4.7 System Features

#	Feature	Description	Priority
1	Secure access	Registration, login, 2FA, role-based permissions	High
2	Menu and stock management	Menu items, time-based menus, daily stock, sold-out status	High
3	Cart, slots and ordering	Cart, pickup slots with capacity, order confirmation with QR code	High
4	Payments, wallet and refunds	Online payment, wallet top-up, automatic refunds, receipts	High
5	Kitchen queue and QR handover	Time-ordered queue, accept / reject, status updates, QR scan	High
6	Notifications and order tracking	Order alerts, pickup reminders, live status	Medium
7	Feedback and reviews	Ratings, comments and manager replies	Medium
8	Dashboards, reports and administration	Sales dashboard, PDF/CSV reports, backup, audit log	Medium

5. ASSUMPTIONS AND CONSTRAINTS

5.1 Assumptions

- Every customer has an institutional email address and a smartphone or computer with a modern browser and internet access.
- The canteen has one kitchen with a fixed number of counters; counter staff have a tablet or PC with a camera for QR scanning.
- The manager sets menu, prices and slot capacity; only one canteen is served in this release.
- Pickup slots are 10 minutes long during canteen hours (7 AM to 7 PM); orders are for same-day pickup only.
- A payment gateway account is provided by the institution or the canteen contractor.
- The institution provides a server or cloud account and an SMTP / SMS service.
- The interface is English only in this release.

5.2 Constraints

Type	Constraint
Technical	Web-based only; open-source stack (PostgreSQL, Linux); must work on current versions of four major browsers; external integrations limited to the payment gateway, SMTP and SMS providers.
Operational	Heavy use in short peaks around breaks and lunch; needs a trained administrator; maintenance only outside canteen hours.
Economic	Budget limited to institutional resources; no paid licences apart from gateway transaction fees; a single small cloud instance in year one.
Legal	Digital Personal Data Protection Act, 2023; card data handled only by a PCI DSS compliant gateway and never stored; veg / non-veg food marking shown on every item; open-source licences respected; WCAG 2.1 AA accessibility.
Time	Prototype in 8 weeks, full release in one semester (about 16 weeks), ready before the next term.
Organisational	Refund, cancellation and no-show rules must match the institution's and the canteen contractor's policy.

6. REQUIREMENT VALIDATION

6.1 Coverage of Stakeholder Requirements

Stakeholder need	Objective	Covered by
Order without queueing	1, 2	FR-11 – FR-16; UC-05, UC-06, UC-18
Live menu and accurate stock	1, 5	FR-07 – FR-11, FR-34; UC-03, UC-04, UC-05
Secure payment and fair refunds	3	FR-17, FR-19 – FR-22; UC-07, UC-08, UC-17, UC-19
Kitchen can manage and fulfil orders	2, 4	FR-23 – FR-27; UC-11, UC-12
Customer is kept informed	1, 4	FR-28, FR-29; UC-09, UC-13
Feedback is recorded and followed up	5	FR-30, FR-31; UC-10, UC-15
Manager gets sales and demand insight	5	FR-32 – FR-34; UC-04, UC-14
Security, availability, auditability	6	FR-35, FR-36; NFR-03 – NFR-07, NFR-12; UC-16

6.2 Quality Checks and Findings

Check	Finding	Improvement made / suggested
Ambiguity	NFR-01 and NFR-02 describe performance with “quickly” and “fast”; FR-14 does not define “capacity” and FR-27 does not define “No-show”.	Each NFR has a measure column as the only test; capacity = maximum orders per 10-minute slot (default 100); No-show = not collected 30 min after the slot ends.
Inconsistency	FR-17 allows cancellation, but FR-24 lets the kitchen accept an order, so it was unclear when cancelling stops.	FR-17 limited to status Received; Cancel is disabled from Preparing onward (UC-08 A1).
Inconsistency	A slot capacity of 20 orders could not serve the 600 orders/hour sized in NFR-09.	Default capacity set to 100 orders per 10-minute slot (about 600 per hour), adjustable by the manager.
Redundancy	FR-25 (order status history) and FR-36 (audit log) both record changes.	Kept: FR-36 is the security and finance trail; FR-25 is the user-visible status, sharing one logging component.
Missing	No allergen or ingredient information for menu items.	Suggested FR-37: manager records allergens and ingredients, shown on the item page.
Missing	No handling of a payment that is deducted but not confirmed by the gateway.	Suggested FR-38: reconcile pending payments through gateway webhooks and auto-refund after 15 minutes.
Missing	Wallet top-up, pickup-slot selection, refunds and password reset were hidden inside other use cases.	Added UC-17 Top Up Wallet, UC-18 Select Pickup Slot, UC-19 Process Refund and UC-20 Reset Password, linked by include / extend.

6.3 Conclusion

Every stakeholder need in Section 1 is traced to at least one functional requirement and one use case. The suggested FR-37 and FR-38 will close the remaining gaps once approved. Every non-functional requirement has a measurable target, and assumptions and constraints are stated. The payment gateway integration (transaction fees and settlement rules) and the 99.5% availability target (institution’s hosting) should be confirmed with stakeholders before design begins.

End of Document — This submission addresses all six required aspects of the Scenario-Based Requirement Analysis: scenario study, functional and non-functional requirements, actors and use cases with diagram, the SRS, assumptions and constraints, and requirement validation.

Rubrics for Calculation

Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Scenario Understanding and Problem Analysis (30%)	Demonstrates thorough understanding of the scenario, stakeholders, objectives, and business context with clear analysis.	Demonstrates good understanding with minor omissions.	Basic understanding with limited analysis.	Poor understanding; major aspects missing or incorrect.
Functional and Non-Functional Requirement Identification (25%)	Clearly identifies comprehensive, accurate, and well-classified functional and non-functional requirements.	Identifies most requirements with minor classification errors.	Identifies only basic requirements; several omissions.	Requirements are incomplete, inaccurate, or poorly classified.
Actor Identification and Use Case Modelling (20%)	Correctly identifies all relevant actors and use cases; use case diagram is complete, accurate, and follows UML standards.	Most actors and use cases identified; diagram has minor errors.	Limited actors/use cases identified; diagram partially correct.	Actors/use cases missing or incorrect; diagram absent or inaccurate.
Software Requirement Specification (SRS) Documentation (15%)	SRS is well-structured, complete, professionally organized, and includes all required sections.	SRS is organized with minor missing sections or formatting issues.	SRS includes only essential sections with limited detail.	SRS is incomplete, poorly organized, or lacks essential components.
Assumptions, Constraints, and Requirement Validation (10%)	Clearly identifies realistic assumptions and constraints; validates requirements for completeness, consistency, and ambiguity with appropriate justification.	Identifies most assumptions and constraints; validation is adequate.	Limited assumptions/constraints; validation is superficial.	Assumptions, constraints, and validation are missing or incorrect.

Criteria	Mark
Scenario Understanding and Problem Analysis (30%)	/5
Functional and Non-Functional Requirement Identification (25%)	/5
Actor Identification and Use Case Modelling (20%)	/5
Software Requirement Specification (SRS) Documentation (15%)	/5
Assumptions, Constraints, and Requirement Validation (10%)	/5
TOTAL	/25